

MPLS L3 VPN Setup:

Thursday, August 13, 2026 11:58 AM

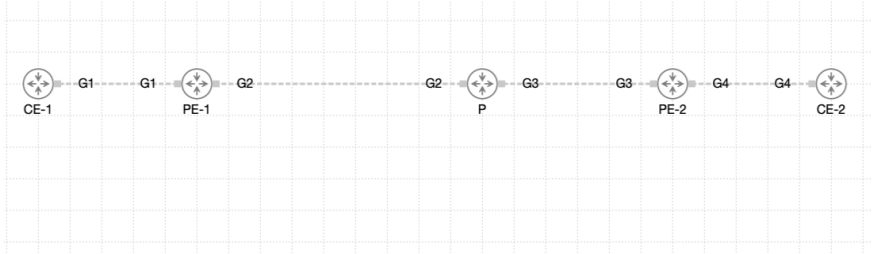
Description: Creating a MPLS L3 VPN setup in Cisco Modeling Labs due to my lab equipment not supporting MP-BGP.

VRF: Porshce

Issue 1 (Fix below under Issue 1 section) : When testing connectivity ran into an issue where both PE-1 and PE-2 had the same ASN for BGP, had to run a "as-override" under BGP address family in the provider network. See below under Issue 1:

Issue 2:

Topology



Configs for CE-1

```
CE-1(config-if)#
CE-1(config-if)#do show run | sec router bgp
router bgp 65000
  bgp log-neighbor-changes
  neighbor 172.20.45.2 remote-as 65001
  !
  address-family ipv4
    network 1.1.1.1 mask 255.255.255.255
    neighbor 172.20.45.2 activate
  exit-address-family
CE-1(config-if)#
```

Interfaces information:

```
CE-1(config-if)#do show ip int br | sec up
GigabitEthernet1    172.20.45.1    YES manual up
Loopback1           1.1.1.1        YES manual up
CE-1(config-if)#
```

```
interface Loopback1
  ip address 1.1.1.1 255.255.255.255
!
interface GigabitEthernet1
  ip address 172.20.45.1 255.255.255.0
  negotiation auto
```

Ce-2 Settings

Interface settings :

```
interface Loopback2
ip address 2.2.2.2 255.255.255.255

interface Loopback3
ip address 3.3.3.3 255.255.255.255

interface GigabitEthernet1
no ip address
shutdown
negotiation auto

interface GigabitEthernet2
no ip address
shutdown
negotiation auto

interface GigabitEthernet3
no ip address
shutdown
negotiation auto

interface GigabitEthernet4
ip address 192.168.1.2 255.255.255.252
negotiation auto
```

```
CE-2#show ip int br | sec up
GigabitEthernet4    192.168.1.2      YES manual up      up
Loopback2           2.2.2.2          YES manual up      up
Loopback3           3.3.3.3          YES manual up      up
CE-2#
```

BGP

```
CE-2#show run | sec router bgp
router bgp 65000
  bgp log-neighbor-changes
  neighbor 192.168.1.1 remote-as 65001
  !
  address-family ipv4
    network 2.2.2.2 mask 255.255.255.255
    network 3.3.3.3 mask 255.255.255.255
    neighbor 192.168.1.1 activate
  exit-address-family
CE-2#
```

PE- 1

Interface settings

```
PE-1(config-router)#do show ip int br | sec up
GigabitEthernet0/1    172.20.45.2      YES manual up      up
Loopback100           100.100.100.100  YES manual up      up
PE-1(config-router)#
```

```

interface Loopback100
 ip address 100.100.100.100 255.255.255.0
!
interface GigabitEthernet0/0
 no ip address
 shutdown
 duplex auto
 speed auto
 media-type rj45
!
interface GigabitEthernet0/1
 vrf forwarding porsche
 ip address 172.20.45.2 255.255.255.0
 duplex auto
 speed auto
 media-type rj45

```

PE-1 VRF Settings

```

PE-1(config-vrf)#
Aug 13 16:43:58.642: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
Aug 13 16:43:59.344: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.do show run | sec vrf
vrf definition porsche
 rd 65001:1
 route-target export 65001:1
 route-target import 65001:1
!
 address-family ipv4
 exit-address-family
 vrf forwarding porsche
PE-1(config-vrf)#

```

ISSUE 1: PE-1 BGP settings : (AS-Override needed to allow see its own ASN from another peer)

```

PE-1>en
PE-1#show run | sec router bgp
router bgp 65001
 bgp router-id 100.100.100.100
 bgp log-neighbor-changes
 neighbor 100.100.100.3 remote-as 65001
 neighbor 100.100.100.3 update-source Loopback100
!
 address-family ipv4
  network 172.20.45.0 mask 255.255.255.252
  neighbor 100.100.100.3 activate
 exit-address-family
!
 address-family vpnv4
  neighbor 100.100.100.3 activate
  neighbor 100.100.100.3 send-community extended
 exit-address-family
!
 address-family ipv4 vrf porsche
  neighbor 172.20.45.1 remote-as 65000
  neighbor 172.20.45.1 activate
  neighbor 172.20.45.1 as-override
 exit-address-family
PE-1#

```

Testing Connectivity

From PE-1 to CE-1 Successful ping

```

PE-1#ping vrf porsche 172.20.45.1

```

```
vrf porsche does not exist% Unable to find vrf 'porsche'

PE-1#ping vrf porsche 172.20.45.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.20.45.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/7/32 ms
PE-1#
```

From CE-1 to PE-1 = Ping successful

```
CE-1(config-router)#do ping 172.20.45.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.20.45.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
CE-1(config-router)#
```

Testing Loopback connectivity from PE-1 to PE-2. Underlay is working fine

```
PE-1#ping 100.100.100.3 source loop100
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.100.100.3, timeout is 2 seconds:
Packet sent with a source address of 100.100.100.100
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
PE-1#
```

Return traffic from PE-2 to PE-1

```
PE-2#ping 100.100.100.100 source loop100
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 100.100.100.100, timeout is 2 seconds:
Packet sent with a source address of 100.100.100.3
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
PE-2#
```

In VRF table Porsche the BGP neighbor is established:

```
Routing Table: porsche
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

 1.0.0.0/32 is subnetted, 1 subnets
B   1.1.1.1 [20/0] via 172.20.45.1, 04:13:55
 2.0.0.0/32 is subnetted, 1 subnets
B   2.2.2.2 [200/0] via 100.100.100.3, 01:00:47
 3.0.0.0/32 is subnetted, 1 subnets
B   3.3.3.3 [200/0] via 100.100.100.3, 01:00:47
 172.20.0.0/16 is variably subnetted, 2 subnets, 2 masks
C   172.20.45.0/24 is directly connected, GigabitEthernet0/1
L   172.20.45.2/32 is directly connected, GigabitEthernet0/1
PE-1#
```

PE-2 settings

Interface settings:

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
PE-2#show ip int br | sec up
GigabitEthernet0/3      10.10.2.2      YES manual up      up
GigabitEthernet0/4      192.168.1.1    YES manual up      up
Loopback100             100.100.100.3  YES manual up      up
PE-2#
```

MP-BGP setting:

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/2/3 ms
PE-2#show ip int br | sec up
GigabitEthernet0/3      10.10.2.2      YES manual up      up
GigabitEthernet0/4      192.168.1.1    YES manual up      up
Loopback100             100.100.100.3  YES manual up      up
PE-2#show run | sec router bgp
router bgp 65001
  bgp router-id 172.20.45.3
  bgp log-neighbor-changes
  network 172.20.45.0 mask 255.255.255.252
  neighbor 100.100.100.100 remote-as 65001
  !
  address-family vpnv4
    neighbor 100.100.100.100 activate
    neighbor 100.100.100.100 send-community extended
  exit-address-family
  !
  address-family ipv4 vrf porsche
    neighbor 192.168.1.2 remote-as 65000
    neighbor 192.168.1.2 activate
    neighbor 192.168.1.2 as-override
  exit-address-family
PE-2#
```

MPLS enabled on all links in the provider network

PE-1- MPLS

```
PE-1#show mpls interfaces
Interface      IP      Tunnel  BGP Static Operational
GigabitEthernet0/2  Yes (ldp)  No      No No      Yes
PE-1#
```

LDP Neighbor :

```
PE-1#show mpls forwarding-table
Local   Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label   Label     or Tunnel Id  Switched     interface
16      No Label  1.1.1.1/32[V] 3420         Gi0/1      172.20.45.1
17      Pop Label 10.10.2.0/30  0            Gi0/2      10.10.1.2
18      16        100.100.100.3/32 0            Gi0/2      10.10.1.2
PE-1#
```

P Router- MPLS:

```
P_Router#show mpls interfaces
Interface      IP      Tunnel  BGP Static Operational
GigabitEthernet0/2  Yes (ldp)  No      No No      Yes
GigabitEthernet0/3  Yes (ldp)  No      No No      Yes
P_Router#
```

```

P_Router#show mpls forwarding-table
Local      Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id Switched      interface
16         Pop Label  100.100.100.3/32 17222        Gi0/3      10.10.2.2
17         Pop Label  100.100.100.100/32 \
                                     17019        Gi0/2      10.10.1.1
P_Router#

```

PE-2 MPLS :

```

PE-2#show mpls interfaces
Interface      IP          Tunnel  BGP Static Operational
GigabitEthernet0/3  Yes (ldp)  No      No  No      Yes
PE-2#

```

MPLS Forwarding Table:

```

PE-2#show mpls forwarding-table
Local      Outgoing  Prefix      Bytes Label  Outgoing  Next Hop
Label      Label      or Tunnel Id Switched      interface
18         Pop Label  10.10.1.0/30 0          Gi0/3      10.10.2.1
19         17      100.100.100.100/32 \
                                     0          Gi0/3      10.10.2.1
20         No Label  2.2.2.2/32[V] 1140       Gi0/4      192.168.1.2
21         No Label  3.3.3.3/32[V] 570        Gi0/4      192.168.1.2
PE-2#

```

Traceroute to verify full connectivity from CE-1 to CE-2

```

Loose, Strict, Record, Timestamp, Verbose[none]:
Type escape sequence to abort.
Tracing the route to 2.2.2.2
VRF info: (vrf in name/id, vrf out name/id)
 1 172.20.45.2 2 msec 1 msec 1 msec
 2 10.10.1.2 2 msec 3 msec 2 msec
 3 192.168.1.1 2 msec 3 msec 2 msec
 4 192.168.1.2 28 msec 3 msec *
CE-1#

```